Machine learning

15-04-26

binary clf model

features $\longrightarrow$ $\boxed{\text{log regr}}$ $\longrightarrow$ $y_{pred} \in [0, 1]$

y_{act}	y_{pred}	
1	0.8	$\rightarrow p_1$
0	0.4	$\rightarrow p_2$
1	0.3	$\rightarrow p_3$

$$\text{BCE loss} = \frac{-1}{3} \sum_{i=1}^{3} y_i \log(p_i) + (1-y_i) \cdot \log(1-p_i)$$

$$= \frac{-1}{3} \left\{ [1 \times \log(0.8) + 0] + [0 + 1 \times \log(0.6)] + [1 \times \log(0.3)] \right\}$$

$$= \frac{-1}{3} [\log(0.8) + \log(0.6) + \log(0.3)]$$

$$\text{BCE loss} = \frac{-1}{n} \sum_{i=1}^{n} y_i \log(p_i) + (1-y_i) \log(1-p_i)$$

$$L = \frac{-1}{n} \sum_{i=1}^{n} y_i \log[\sigma(z)] + (1-y_i) \log[1-\sigma(z)]$$

$$z = w_1 x_1 + w_2 x_2 + b$$

here the reln between loss and model parameters is convex

Due to given convex reln, we can minimize BCE loss by finding optimal values of model parameters using <u>gradient</u> <u>descent</u>.

$$L = f(w, b)$$

$$w_{new} = w_{old} - \alpha \frac{\partial L}{\partial w_{old}}$$

$$b_{new} = b_{old} - \alpha \frac{\partial L}{\partial b_{old}}$$

$$L = -\frac{1}{n} \sum_{i=1}^{n} y_i \log(p_i) + (1-y_i)[1-\log(p_i)]$$

put $t = \log(p_i)$

$p_i = \sigma(z)$

$z = w_1 x_1 + w_2 x_2 + b$

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial t} \cdot \frac{\partial t}{\partial p} \cdot \frac{\partial p}{\partial z} \cdot \frac{\partial z}{\partial w_1}$$

$$L = -\frac{1}{n} \sum_{i=1}^{n} y_i t_i + (1-y_i)(1-t_i)$$

$$\frac{\partial L}{\partial t_i} = -\frac{1}{n} \sum_{i=1}^{n} \frac{\partial y_i t_i}{\partial t_i} + \frac{\partial (1-y_i)(-t_i)}{\partial t_i}$$

$$= -\frac{1}{n} \sum_{i=1}^{n} y_i + y_i - 1$$

$$\frac{\partial L}{\partial t_i} = -\frac{1}{n} \sum_{i=1}^{n} 2y_i - 1$$

$t = \log(p)$

$y = \log(x)$

$$\frac{\partial y}{\partial x} = \frac{1}{x}$$

$$\frac{\partial t}{\partial p} = \frac{1}{p}$$

$p = \sigma(z)$

$$\frac{\partial p}{\partial z} \checkmark$$

$$\frac{\partial z}{\partial w_1} = \frac{\partial (w_1 x_1 + w_2 x_2 + b)}{\partial w_1}$$

$$\frac{\partial z}{\partial w_1} = x_1$$

$$\boxed{\frac{\partial L}{\partial w_1} = \frac{1}{n} \sum_{i=1}^{n} (p_i - y_i) x_{i1}}$$

$$\frac{\partial L}{\partial w_1} = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i) x_{i1}$$

$$\hat{y}_i = p_i = \sigma(z_i) = \sigma(w_1 x_{i1} + w_2 x_{i2} + b)$$

$$\boxed{\frac{\partial L}{\partial b} = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)}$$

training steps

1) random initialization of model parameters

2) compute loss and gradients $\frac{\partial L}{\partial w}$ $\frac{\partial L}{\partial b}$ at current values of w and b

3) apply parameter updation

$$w_{new} = w_{old} - \alpha \frac{\partial L}{\partial w_{old}}$$

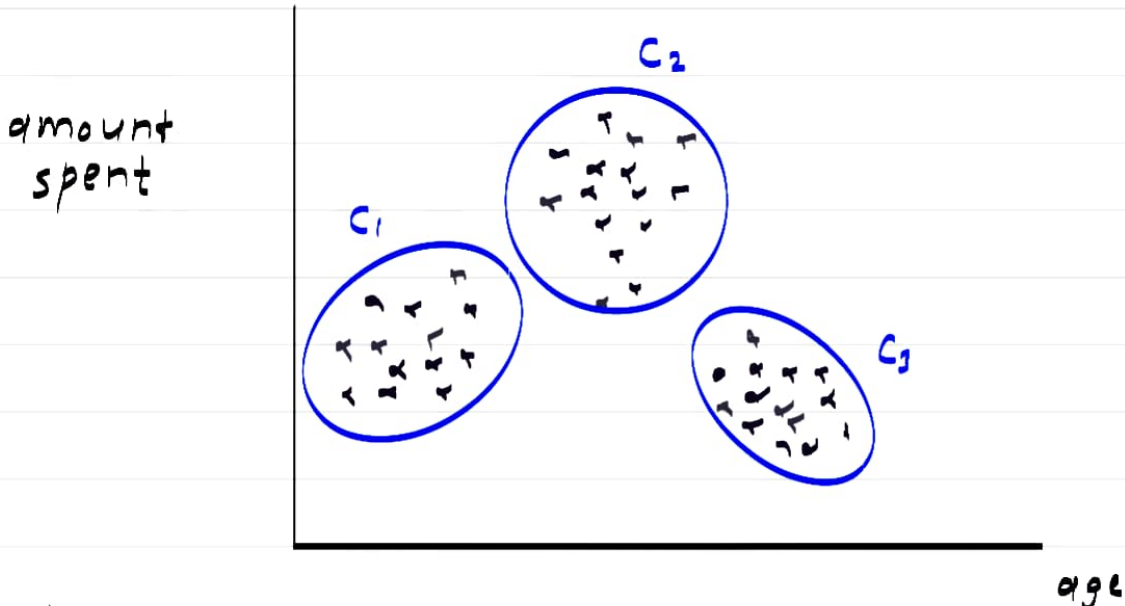
$$b_{new} = b_{old} - \alpha \frac{\partial L}{\partial b_{old}}$$

4) repeat steps 2 and 3 untill convergence

KMeans Clustering

- unsupervised ML algorithm
- works on unlabelled data
- distance based algorithm

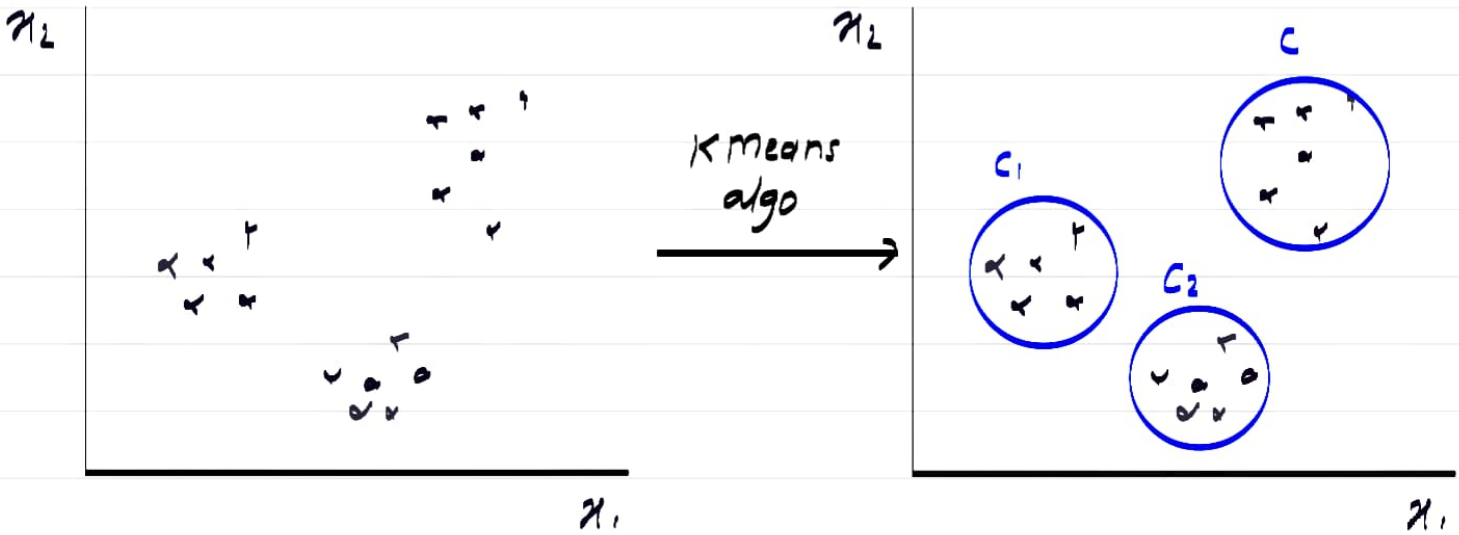
eg. customer clustering/ segmentation



- customer segmentation is very important in business
- it helps in delivering better customer experience

Objective of KMeans Clustering:

KMeans algo tries to capture hidden patterns in features to find relevant clusters

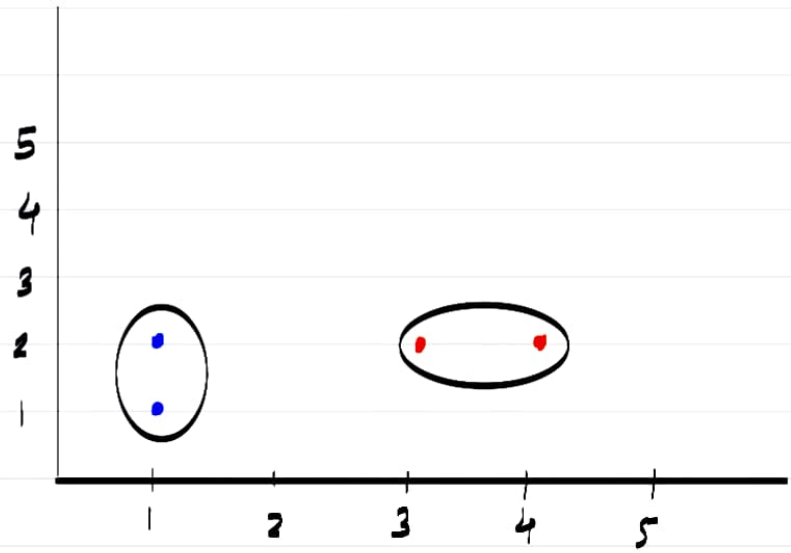


once model is trained

new datapoint x_1, x_2 → $\boxed{\text{KMeans clust}}$ → C_1, C_2, C_3

Training process of KMeans clustering :-

eg.	x_1	x_2
d_1	1	1
d_2	1	2
d_3	3	2
d_4	4	2



Step 1 - decide value of K - no. of clusters to find

in above example lets say $K = 2 \begin{cases} C_1 \\ C_2 \end{cases}$ 2 clusters

K is a hyperparameter which needs hyperparameter tuning

Step 2 - randomly assign K centroids for K clusters

for $K = 2$

randomly pick 2 datapoints from data and assign them as centroids

$C_1 = d_1 = (1, 1)$

$C_2 = d_2 = (1, 2$